

Programmer's Model of 8086

Q. Describe the programmer's model of the 8086 microprocessor. Detail the General Purpose Registers, Segment Registers, Pointer/Index Registers, and the Flag Register.

> **Definition to Remember**

> The
divided

1. General Purpose Registers (16-bit / 8-bit split)

Can be used as one 16-bit register or two 8-bit registers (High/Low).

- **A**

2. Segment Registers (16-bit)

Hold the base addresses of 64 KB memory segments to help address 1 MB of physical memory.

- **C**

3. Pointer and Index Registers (16-bit)

Hold offset addresses relative to the segment registers.

- **S**

4. Flag Register (16-bit)

Contains 9 active flags (6 Status, 3 Control).

- **S**

> **Must-Write Points (for 10 marks)**

> 1. C

> **Quick Recall**

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Contro